Usability Evaluation of the Aging Scenario Wiki as a Learning Tool

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Abstract. The Chinese government has developed a unique "Smart Eldercare" model that utilizes digital technologies to address the growing needs of its rapidly aging population. To implement this model, the government has initiated policies to accelerate talent cultivation, highlighting the importance of Smart Eldercare education at different levels in universities and vocational schools. This paper presents the design and development of an aging scenario Wiki and conducts a usability evaluation of this Wiki as a collaborative learning tool for nascent Smart Eldercare education. First, during the development phase, we conducted lab-based usability testing using 18 tasks of 3 types with 12 student participants who followed the think-aloud protocol. Our analysis identified seven usability issues, and we subsequently addressed them in the optimization phase. Second, we evaluated the usability of the aging scenario Wiki as a learning tool in actual work teaching activities involving 54 students who collaborated on creating aging scenarios. The evaluation yields a System Usability Scale score of 67.45 and a C usability rating, indicating the areas for further improvement. Overall, the aging scenario Wiki demonstrated the potential to be a practical and effective learning tool for nascent Smart Eldercare education. Our study highlights the importance of user-centered design and usability testing in developing educational resources for emerging fields such as Smart Eldercare.

Keywords: Aging Scenario · Wiki · Usability Testing · Collaborative Learning · Smart Eldercare

1 Introduction

1.1 Background and Motivation

With the aging global population, China is exploring a unique "Smart Eldercare" model to respond to the social service needs of older adults [8]. This approach integrates information technology with eldercare services to offer intelligent care solutions for older adults. This model represents a cutting-edge concept in eldercare.

Education is an effective means to disseminate the concept of Smart Eldercare. Smart Eldercare education can help the public form the proper perspective of aging to encourage the integration of eldercare and new technologies. It helps provide intelligent solutions to eldercare services for older adults.

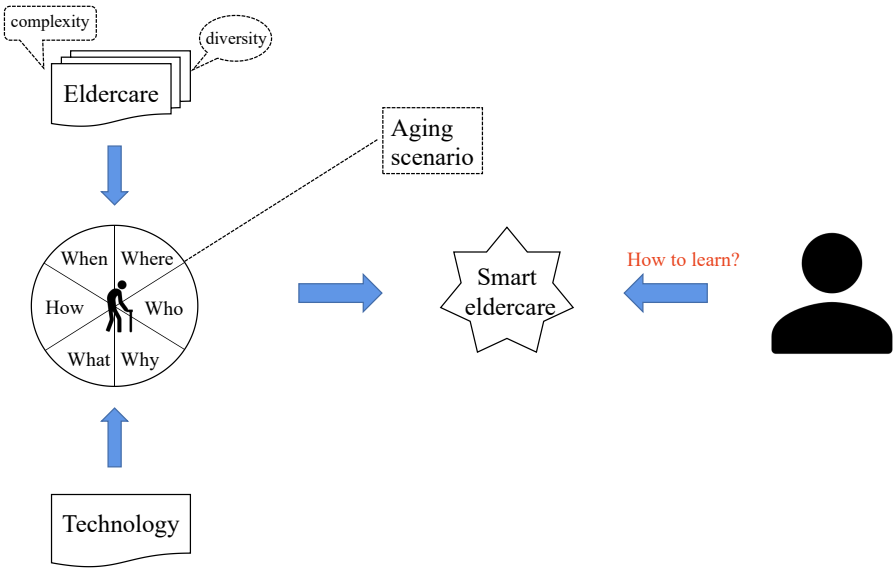


Fig. 1. The challenge of Smart Eldercare education

1.2 Objectives and Research Questions

In recent years, the higher education system has begun to introduce eldercare courses. Some colleges of pharmacy offer geriatrics-focused elective courses to train healthcare professionals [14]. To enhance pharmacotherapy education in geriatrics, Estus et al. have proposed an active-learning course model that involves site visits, interviews, and interactions with older adults, leading to a positive perception of aging among students. Experiential geriatric education

has also been integrated into pharmacy curricula in Qatar and Canada, aiding in students' preparation for eldercare [32]. In some universities, geriatric education is included in the health curriculum, requiring students to partner with elderly individuals for people-centered nursing, which has yielded favorable results [30]. However, the majority of the current eldercare education adopts teacher-centered learning which ignores teaching students how to learn and encouraging lifelong learning. How to use student-centered learning to encourage students to learn the knowledge of Smart Eldercare autonomously and develop active methodologies remains a challenge.

1.3 Significance and Contributions

This study developed an aging scenario Wiki as a collaborative learning tool for nascent Smart Eldercare education. We incorporated the elements of When, Where, Who, Which, What, and How into the aging scenario to present the demands of older adults intuitively and effectively. During the construction of the Wiki, we designed and conducted a lab-based usability test, optimizing the website based on the test results. Further, we integrated the Wiki into practical teaching activities that involved 54 students collaborating to develop aging scenarios. Subsequently, we assessed the Wiki's usability as a learning tool for this course using the SUS score, which resulted in a C-grade usability rating with a score of 67.45.

2 Related Works

2.1 Scenario as a Teaching Medium

The term "scenario" finds its root in literature and drama, originally referring to the specific character's behavior and actions in a certain time and space. Joshua Meyrowitz elaborated the meaning of scenario as: "a specific place, in most cases, includes specific people, specific time and specific activities [19]." These descriptions show that "scenario" is the organic unity of people, environment, needs, and activities, capturing the life picture of a specific person taking certain actions to meet a certain demand in a specific time and space.

In recent years, the concept of "Scenario" has been extended beyond its original definition and has become a mode of thinking that connects products and users. Scenarios allow those who need information and those who provide it to gain a shared understanding through interaction [22]. Furthermore, scenarios are useful for describing and evaluating situations, expressing needs, and providing opportunities to understand intentions and information [16]. The application of scenarios in the development of AI applications for nursing care is seen as promising by Seibert et al. [26]. Several governments have begun constructing Smart Eldercare application scenarios, stimulating the development of the Smart Eldercare industry, and improving the acquisition and satisfaction of older adults. The research on different aging scenarios helps to pay attention to the actual needs of older adults and improve the eldercare service [1].

2.2 Collaborative Learning and Knowledge Sharing

Collaborative learning [28] is an educational method to teach and learn that involves groups of learners working together to solve problems, complete tasks, or create products. It also involves a community of learners and teachers where members acquire and share experiences or knowledge [31].

The goal of collaborative learning is to guide the participants to build their knowledge through exploration, discussion, negotiation, and debate [31]. In collaborative learning, the core elements are positive interdependence, individual responsibility, face-to-face and verbal interaction, social skills, and team reflection. Johnson et al. think of collaborative learning as an interactive system carefully designed to organize and lead to interaction among team members [12]. This way, each member of the team is responsible for their own learning as well as that of the rest of the group. Salinas argues that the acquisition of a range of competencies and attitudes is a result of group interaction [24], while Barros et al. insist that it is a process by which the students learn as they suggest and share ideas to solve the task [4]. Collaborative learning arises from the interaction between individuals. Based on their respective contributions, each member creates new knowledge and develops capabilities that benefit other participants.

2.3 Usability Testing in Human-Computer Interaction Research

Usability testing is a process that allows actual users to test a product and collect user feedback [23]. The formative and summative conceptions of usability testing serve different purposes. The former aims to identify and address usability issues during the product development process to enhance its usability [23], while the latter focuses on measuring the extent to which users can achieve their goals using the product [11]. Before each test, it is necessary to clarify the specific objectives before formulating the test plan.

Usability testing can be divided into laboratory testing and remote testing according to the different environments of test users. Both types of tests can capture the main usability issues of a website [29]. Laboratory-based usability testing is a traditional way. It is effective in revealing the usability issues of websites and other applications [5, 6]. A typical laboratory-based usability test involves relatively few participants [21]. Participants represent real users, come to the laboratory alone, and use the system or website for fundamental tasks. Observers in the same room or adjacent rooms usually record the time required for users to complete each task, the completion of the task, and what participants do and say while performing the task. At the end of the task, there are two broad classes of usability test results: problem reports and quantitative measurements. Problem reports mean describing usability problems that prevent a target user from achieving a target task with reasonable effort and within a reasonable time [18]. The quantitative measurements use standardized usability questionnaires. The most widely used standardized usability questionnaires are Questionnaire for User Interaction Satisfaction [9], the Software Usability Measurement Inventory [13], the Post-Study System Usability Questionnaire [15], and the System Usability Scale [7].

3 Design and Development of Aging Scenario Wiki

In this section, we developed an Aging Scenario Wiki for Smart Eldercare education. Fig 2 shows the architecture.

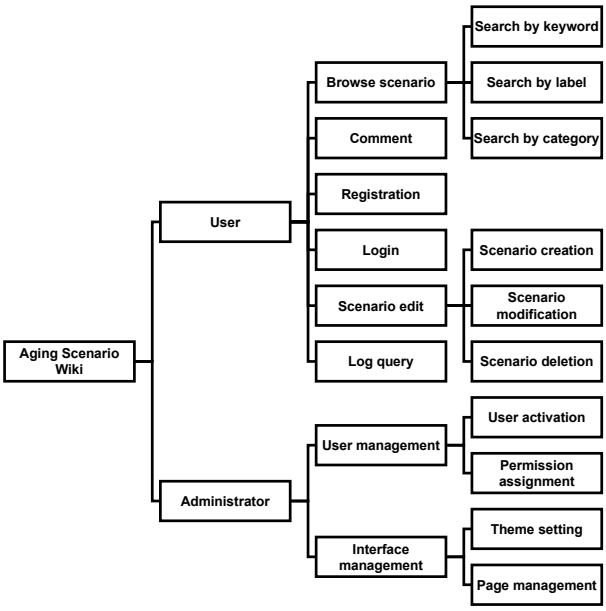


Fig. 2. The architecture of Aging Scenario Wiki

3.1 Scenario-based Approach to Smart Eldercare Education

We present a teaching approach for incorporating aging scenarios into Smart Eldercare education. We use the aging scenario to present the demands of older adults. Each aging scenario includes a set of nine entries [1]. They are background, scenario elements(including main users, main places, main people, occurrence frequency, impact depth, impact breadth, and urgency), scenario cases, scenario requirements, existing schemes, emerging schemes, associated scenarios, scenario labels, and references.

We engaged two undergraduate students and seven postgraduate students from Hefei University of Technology to manually create 140 aging scenarios over a period of 15 days. From the perspective of older adults, based on Maslow's hierarchy of needs, we divided five hierarchies of needs for physiology, safety, belongingness and love, esteem, self -actualization. We have also considered caregivers, communities, eldercare institutes, government, and society to include more perspectives. Under each hierarchy of needs, we set up different categories. See Tab 1

and Tab 2 for the specific number of scenarios in each category. These scenarios will become the first batch of knowledge in the aging scenario Wiki.

Table 1. The perspective of the older adults

Maslow's hierarchy of needs	**Frequency**	**Categories**	**Frequency**
Physiological needs	17	Air	1
		Water	1
		Food	2
		Sleep	3
		Transportation	7
		Excretion	1
		Sex	2
Safety needs	61	Safety	25
		Safety	21
		Employment	3
		Property	8
		Resource	4
Belongingness and love needs	18	Friendship	1
		Family	2
		Affiliation	1
		Emotion	14
Esteem needs	8	Self-esteem	6
		Confidence	2
Self-actualization needs	10	Creativity	1
		Release potential abilities	4
		Accept the fact	1
		Spontaneity	4

3.2 Wiki as a Collaborative Learning Tool for Smart Eldercare Education

Wiki is a tool based on Web 2.0 technology. Web 2.0 is a user-centered Internet application model that promotes collaboration and communication between people by allowing users to create, share, and interact with content. Wiki allows users to create and edit content and constantly improve and update it through collaboration. It can enable users to focus on information exchange and collaboration tasks, and they will not be disturbed by the complex technical environment [2, 17]. Users can use the Wiki to document work and continuously record iterations when developing projects. In particular, wiki users can collaborate to develop topics jointly. Also, the wiki environment can record the individual changes in the system at each time to restore the page to its previous state at any time. As a tool for group authoring, wikis can pull team members into a group, allowing them to jointly edit and build documents on one interface [10]. At the

Table 2. Hierarchy of needs under multiple perspectives

Hierarchy of needs	Frequency	Categories	Frequency
Caregiver needs	5	Caregiver	5
Community needs	6	Community infrastructure	1
		Community management and construction	5
Eldercare institute needs	9	Management of eldercare institutes	6
		Cognitive disorder management in eldercare institutes	2
		Operation in eldercare institutes	1
Government needs	1	Government regulation	5
Society needs	1	Society	5

same time, by continuously acquiring new knowledge and the maintenance of administrators, the Wiki can become a knowledge base focusing on a certain kind of knowledge. In a broad sense, a wiki can be considered a learning object if it contains an educational purpose [20].

3.3 Development of Aging Scenario Wiki

We deploy the aging scenario Wiki on wiki.js (version No. 2.5.219). The operating system is a Linux system based on a docker container, with a 4-core CPU and 15.67GB memory allocated. The database uses PostgreSQL (version No. 13.3), and the running environment is based on node.js (version 14.17.6). The Wiki is an open platform, and the URL is https://stgawiki.dataee.net.

As shown in Fig 2, the Wiki defines administrator and user.

Adminstrator The administrator is a built-in user and can manage users, including user verification and permission assignment. After user verification, the administrator can assign users to groups with different levels of permissions. Additionally, administrators have interface management capabilities and can change the website's theme style. As a superuser, the administrator has the ability to view and modify all pages within the wiki.

User There are three types of users: guest, student, and instructor.

Any visitor can visit the website as a guest. The aging scenario Wiki is designed to be open to the public, allowing visitors to browse and comment on aging scenarios without the need for registration or login. The website provides a search function to facilitate users in finding and locating aging scenarios. The search function can search based on keywords, tags, or categories provided by the user, and display results that match the search criteria on the website.

The instructor needs to register and wait for the administrator's verification to get an account with the instructor's permission. The instructor can create a Smart Eldercare class wiki and invite students to join the course. In the class wiki, the instructor has the permission of an administrator.

Upon registering and logging in to the website, a student is invited by the instructor to the Smart Eldercare class. The class features an operation guide for students to reference, followed by the ability to create and edit pages within the class wiki. Notably, each page has a page creator, and two other features: visibility and modifiability. The visibility and modifiability of a page can only be authorized by the page creator.

In the course, both the instructor and students have the opportunity to fully leverage the wiki's functionality to facilitate course-related activities. The instructor can generate a page containing course materials, announcements, and other relevant information. Students are able to edit their own notes, class discussions, and assignments on their respective pages. When working in groups, students can create a discussion page to exchange ideas and collaborate on developing projects.

After the course, the instructor is able to utilize the log function to evaluate the usage of the Wiki by students. The log tracks various activities, such as visits to others' wikis, comments, collaborative editing, and discussions. They can be used for the evaluation of course grades. The final submitted assignment can also be used as a scoring item. If it is a personal assignment, the instructor can score it directly. If it is a collaborative writing task, the instructor can view their specific contribution through the log and avoid laziness in the group.

4 Usability Testing of Aging Scenario Wiki

In this section, we designed a usability test and optimized the wiki site.

4.1 Lab-based Usability Testing Methodology

Setting the Goals The goal of the usability test was to confirm how the aging scenario Wiki works when used by users. To further solve this problem, the research process was divided into three tasks. The three tasks were the guest task, the student task, and the instructor task.

Designing the Test With the clear test goal, we formulated specific test questions representing the above roles. According to the different roles of users, we developed three different types of tasks. The questions under each type of task were to test the design of the website, not the skills of users.

We first conducted a pilot test with a novice student. Based on the results of the pilot test, we identified a need to slightly modify the wording of several questions to eliminate ambiguity and facilitate users' understanding. We also eliminated two redundant questions and rearranged the sequence of the questions in the test questionnaire. As a result of these modifications, the final version of the usability test questionnaire consisted of 18 task-oriented questions.

guest task

1. Click on a scenario of interest on the home page and read it.
2. Write comments under the article for the first task.
3. Does the website have the scenario of older adults falling in the bathroom?
4. Is the scenario of kinship maintenance under the Family Category?
5. Find some safety-related scenarios of older adults through label search.
6. How do you find a scenario similar to one of the older adults falling in the bathroom?

student task

1. Is there an operation guide on the website?
2. Create a scenario of emergency rescue for the older adult (XX) in the class wiki, XX is your test ID, and the template is the scenario of emergency rescue for the older adult.
3. Add the picture_ (33).png under the photo file for case one of the scenario of emergency rescue for the older adult.
4. Change the scenario demand of the emergency rescue for the older adult to "if the elderly have abnormal physical conditions when they are alone at home..."
5. Add the scenario of prevention of getting lost for the older adult with home cognitive impairment in the associated scenario and add hyperlinks.
6. Delete the fourth reference of the scenario of emergency rescue for the older adult.

instructor task

1. Create a course (XX) wiki, XX is your test ID.
2. Check the scenario of sleep health of the older adult (XX). Whether there is any unreasonable place in the scenario. If so, inform the contributor to modify it. If not, skip the task.
3. Query the log of the above scenario.
4. Move the location of the scenario of sleep health of the older adult (XX) to the course (XX) wiki.
5. Delete the scenario of the safety of caregivers (XX).
6. Contact the administrator.

After the final confirmation of the test questions, we selected problem reports to present the test results. The test results focused on identifying usability issues with the Wiki site rather than providing a detailed analysis of its performance. We considered complex data collection techniques unnecessary. Instead, We use the think-aloud method [27], which involved the participants explaining their thought processes and opinions and interacting with a tester to complete tasks. This allowed us to gain direct insight into which parts of the interface caused the most problems.

Finally, we outlined the details of our usability test design, including the duration, required equipment, and testing environment. We planned to conduct tests at regular intervals of every half hour and complete all tests within two days. Participants were required to have a standard personal computer with internet access and a web browser. To provide a relaxed and realistic testing environment, we allowed participants to use their own computers. The only limitation was that the test environment should be quiet and private enough. Therefore, we prepared a dedicated room that was soundproof and comfortable for participants to use, which did not interfere with their testing experience.

4.2 Participants and Tasks

We recruited 3 female graduate students and 9 male graduate students from the Hefei University of Technology to participate in the usability test of the website.

In the study, we performed 12 rounds of testing, each with one participant and one tester. Each participant would be informed to complete 18 task-oriented questions. The tester was responsible for interacting with participants and recording what happened during the test. Before the task, participants received a questionnaire containing 18 questions and a test ID that logs in to the system. Each participant was asked to read the question aloud and describes their thoughts using the think-aloud method when completing the task and browsing the website.

Each usability test lasted approximately 40 to 60 minutes. The tester encouraged participants to attempt as many questions as possible. We redirected the user's initial access interface to the home page, allowing participants to directly perform the first task after logging into their account. They were asked to complete the tasks in order. But if they failed to find an answer to a question, they could choose to skip the question or ask the tester for help. Before the test ends, the tester would ask the participant if they were ready to end the test. After receiving an affirmative answer, the tester announced the end of the test.

After the usability test, participants were asked to provide feedback, including their opinions on the website. The tester then reviewed the recorded observations and comments to identify issues with the website and functions that worked well.

4.3 Results Analysis and Optimization

The results of the usability test showed that the participants were able to complete all the tasks, indicating that the website's predefined functions were realized. Meanwhile, we identified a total of 7 usability issues from the 12 observer records. In response, we made some changes to the aging scenario Wiki.

During the test, participants used their commonly used screen resolutions, with most using 1366 * 768 or 1600 * 900 pixels. However, due to the home page layout being designed for a 1920 * 1080 pixel screen, most participants experienced incomplete page content display. To address this issue, we reconfigured

the CSS of the web page and set different page layouts for users with different computer resolutions.

When reading the aging scenario, some participants raised doubts about the three elements of users, places, and people in the scenario. They could not understand the meaning of ticking the options before them. Participants thought that only the relevant options should be added and unsuitable options could be removed. We considered this suggestion reasonable and adopted it.

During testing, the participants found the guidance on the home page to be unclear. Specifically, when searching for the operation guide, 50% of participants initially overlooked the "operation guide" link on the home page. Instead, they looked for the operation guide through the navigation bar or search box. It took them longer. Participants noted that the font size was too small and the font color was too similar to the background, making it difficult to notice. Some participants suggested changing the font color to white, which we implemented to improve visibility.

Participants faced difficulties when using the navigation bar, stating that it was cluttered and lacked hierarchy, making it inconvenient to locate scenarios by category. To address this issue, we streamlined the navigation bar, reducing it to five options: home page, label, user center, operation guide, and contact administrator. Additionally, the categories are now located in the root directory, enhancing the user experience.

Searching by labels is an important method for finding information. But during our usability testing, 75 percent of the participants could not locate the label search function on their first attempt. They felt that the button for label search was not prominent and the location was too remote. Once finding the label button, the participants found that the distribution of labels was disorganized and difficult to navigate, with redundant and unnecessary labels. To address these issues, we streamlined the scenario labels and redesigned the label search button to make it more prominent and accessible.

When performing the scenario construction task, the customize "template" option did not meet the usability standard. Participants did not realize that the "template" can help them quickly build a scenario architecture. Instead, they chose to create a new blank interface and built the scenario from scratch. Only a few participants used this option. As a result, most participants spent a lot of time completing this task, and the results were not satisfactory. Therefore, we described the role of "template" in detail in the "operation guide" so that the users could understand the function of the "template" button.

Participants tended to ignore the administrator's contact information at the bottom of the home page when finding the administrator's contact information. Instead, they relied on their previous experience to locate the answer in the settings. To make it easier for users to provide feedback and suggestions, the administrator's contact information has been moved to a more prominent location.



Fig. 3. The interface of an aging scenario Wiki

In summary, the observation of testers and the feedback of students participating in the test are valuable in revealing the causes of site failure and helping find the usability issues. Fig 3 shows an aging scenario interface after iteration.

5 Evaluation of Aging Scenario Wiki as a Learning Tool

In this section, we used the Wiki in actual teaching activities and evaluated its usability as a learning tool.

5.1 Use of Aging Scenario Wiki in Actual Teaching Activities

Fig 4 shows a group of teachers and students carrying out a Smart Eldercare course, by using a wiki, https://stgawiki.dataee.net/. The program was intended to deepen students' understanding of aging scenarios through collaborative learning. The course restricted only students could publish aging scenarios on the platform, while it was a public wiki that could be freely accessed by the public.

Participants 54 second-year students majoring in geriatric health care and management from Anhui Medical College participated in the teaching activities, comprising 24 male and 30 female students.



Fig. 4. Using aging scenario Wiki in a Smart Eldercare course

Procedure Before the commencement of the course, we invited a teacher and assigned her instructor permission to utilize our Wiki and negotiated this teaching activity.

During the teaching activity, the teacher first explained the activity. The students were told that their task would be to work in a team to develop wiki entries to elaborate on the concepts or descriptions of aging scenarios. They were assigned this task and provided a 120-minute deadline to complete it. At the same time, the teacher informed the students that the platform kept a history of the aging scenarios development process so that teachers could evaluate each student's contribution. The feature helped the teacher to evaluate the work of each group member. In addition, the teacher was responsible for clarifying any concerns with the project and the use of the wiki platform. The teacher guided the students to ensure their work would achieve the requisite quality.

Then, the students started the development of the aging scenarios. They were divided into 9 groups. After familiar with the operation of the Wiki through the operation guidelines we provided them, they logged in to the Wiki and uploaded aging scenarios about the digital divide designed before the class. The teacher showed some existing problems with the uploaded aging scenarios and instructed students to use several search tools which could help them during the development of the project. From that point on, the groups collaborated to iterator the aging scenarios with the team. The creator of the aging scenario allowed the rest of the group to read the page and contribute as necessary. The group members searched for information about the aging scenario they collaborated on editing to help the creator iterate the aging scenario. When encountering difficulties, the team members sought solutions through active dialogue and communication.

At the end of the course, we invited students to take a questionnaire about our aging scenario wiki site. The online questionnaire aimed to evaluate the usability of using our Wiki as a learning tool for Smart Eldercare teaching.

5.2 Student Feedback and Evaluation

During the course, 54 students majoring in geriatric health care and management developed and contributed 150 aging scenarios related to the digital divide to the wiki platform used in the teaching activities. These scenarios covered a wide range of issues, such as social isolation among older adults, access to healthcare services, digital literacy, and technology acceptance.

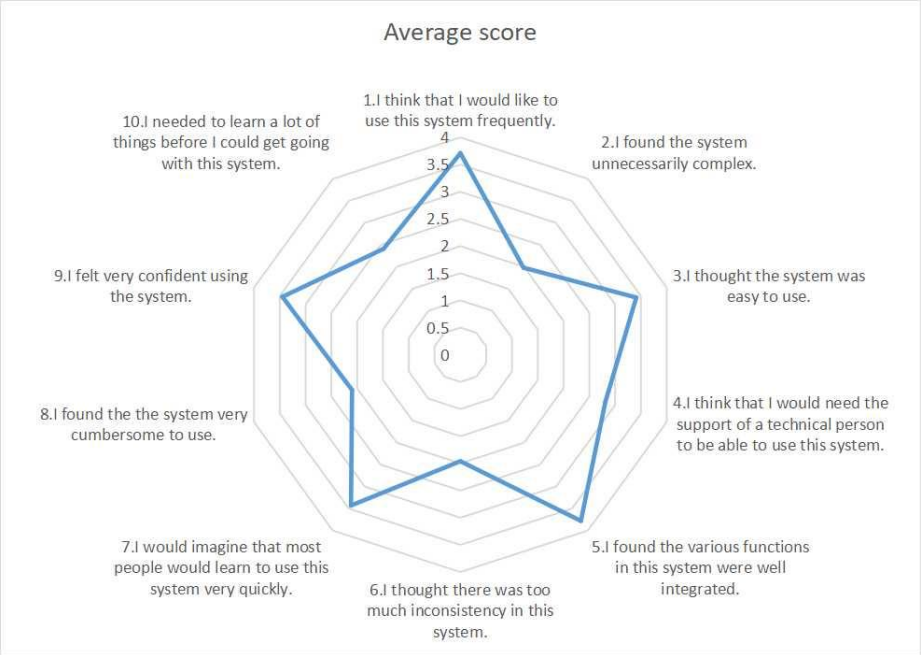


Fig. 5. The questionnaire score

We evaluated the students' perceptions and satisfaction through System Usability Scale [7]. It consists of 10 items, odd-numbered items are positive statements, and even-numbered items are negative statements. It is divided into completely disagree, disagree, unsure, agree, and strongly agree according to the perceived intensity of the audience, corresponding to a score of 1-5, respectively. SUS scores reflect overall satisfaction, while questions 4 and 10 measure learnability and the other 8 questions measure usability [3]. We collected survey data through 53 questionnaires.

Fig 5 and Tab 3 show the results. From Tab 3, the SUS score is 67.45. According to the SUS database [25], it can be converted to a percentile rank which means that the Wiki's usability is higher than about 50 percent of other websites. According to the SUS grading range of the fractional curve [3], the usability rating of this system is C and the adjective rating is "OK". Questions 1, 6, and 9 show high user satisfaction and students are more inclined to use the Wiki as a tool for lifelong learning.

According to Fig 5, we find that the learnability score is only 60.38, indicating that our Wiki requires a certain learning cost or time for first-time users. Based on feedback from students, we learn that most of the difficulties occur when creating pages. They need guidance from professional technical personnel or a detailed operation manual to help use the Wiki for the first time. On the other side, the usability score is 69.22 which shows the usability of the Wiki is acceptable.

Table 3. SUS score

SUS score	Learnability score	Usability score
67.45	60.38	69.22

6 Discussion and Conclusion

This study builds an aging scenario Wiki as a collaborative learning tool for nascent Smart Eldercare education. The Wiki site focuses on older Chinese adults and uses scenarios as the medium of knowledge exchange. We conducted a user-centered two-stage usability evaluation on it.

Firstly, we conducted a formative evaluation to identify potential usability issues. We designed three types of 18 tasks, recruited 12 testers, and tested the usability of the Wiki site during its construction. We chose problem reports to present the test results. According to the feedback from the think-aloud protocol, we identified 7 usability issues and optimized the Wiki site accordingly.

Secondly, we performed user testing with a group of 54 participants to assess the system's usability in a more realistic setting. We incorporated it into our teaching activities, and 54 students actively engaged in developing 150 aging scenarios about the digital divide. The development of these aging scenarios was a collaborative effort among the students, who worked in groups to identify and analyze the challenges faced by older adults in using digital technologies. They then used this analysis to develop scenarios that highlighted the potential risks and benefits of technology use in the context of aging. After the course, we evaluated the usability of Wiki as a learning tool through quantitative measurements using SUS. The SUS score was 67.45 and the usability rating of the Wiki was C.

Overall, this Wiki provided the students with a practical and meaningful learning experience that allowed them to apply their knowledge and skills to real-world problems in the field of Smart Eldercare. Our study emphasizes the significance of adopting a user-centered design in developing educational resources for emerging fields such as Smart Eldercare.

One limitation of this study is the C-grade usability rating of the Wiki indicated by the SUS score. It indicates the Wiki has the area for further iterative improvement. Future research will focus on improving the usability of the Wiki platform. This could involve conducting additional user testing and collecting feedback from users to identify specific areas for improvement. The iterative design process can then be used to implement changes and improvements to the platform.

Acknowledgements. This work was partially supported by Anhui Province Higher Education Institution Quality Project "Smart Eldercare" (2019kfkc010) and Hefei University of Technology Teaching Reform Demonstration Course Project "Smart Eldercare" (KCSZ2020038).

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